

Fall 2018

Problem 1 (Fa08, Fa18). Show that $\int_0^1 \frac{1}{x^x} dx = \sum_{n=1}^{\infty} \frac{1}{n^n}$.

Problem 2 (Sp77, Su82, Fa18, Fa21). Suppose f is a differentiable real valued function such that $f'(x) > f(x)$ for all $x \in \mathbb{R}$ and $f(0) = 0$. Prove that $f(x) > 0$ for all positive x .

Problem 3 (Fa18). Let X be a metric space.

1. If U is a subset of X show that there is a unique open set $\text{ext}(U)$ disjoint from U and containing all open sets disjoint from U .
2. Give an example of an open set U with $U \neq \text{ext}(\text{ext}(U))$.
3. Prove that for all open sets U , $\text{ext}(U) = \text{ext}(\text{ext}(\text{ext}(U)))$.

Problem 4 (Fa18). Let a be a real number with $|a| < 1$. Prove that

$$\sum_{k=1}^{\infty} a^k \cos k\theta = \frac{-a^2 + a \cos \theta}{1 + a^2 - 2a \cos \theta}$$

Problem 5 (Fa18). Describe a conformal map from the set $\{|z - 4i| < 4\} \cap \{|z - i| > 1\}$ onto the open unit disc.

Problem 6 (Fa05, Fa18). Let A be an $n \times n$ matrix with real entries such that $(A - I)^m = 0$ for some $m \geq 1$. Prove that there exists an $n \times n$ matrix B with real entries such that $B^2 = A$.

Problem 7 (Fa18). Suppose $A = (a_{ij})$ is a real symmetric $n \times n$ matrix with nonnegative eigenvalues. Show that

$$|a_{ij}| \leq \sqrt{a_{ii}a_{jj}}$$

for all distinct $i, j \leq n$.

Problem 8 (Fa84, Fa96, Fa10, Fa18). Let gcd abbreviate greatest common divisor and lcm abbreviate least common multiple. For three nonzero integers a, b, c , show that

$$\gcd\{a, \text{lcm}\{b, c\}\} = \text{lcm}\{\gcd\{a, b\}, \gcd\{a, c\}\}.$$

Problem 9 (Fa18). Suppose a prime number p divides the order of a finite group G . Prove the existence of an element $g \in G$ of order p .

Problem 10 (Fa18). A mathematician (stupidly) tries to estimate

$$\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}$$

by taking the sum of the first N terms of the series. What is the smallest value of N such that the error of this approximation is at most 10^{-6} ?

Problem 11 (Su84, Fa18). Let $p(z)$ be a nonconstant polynomial with real coefficients such that for some real number a , $p(a) \neq 0$ but $p'(a) = p''(a) = 0$. Prove that the equation $p(z) = 0$ has a nonreal root.

Problem 12 (Su83, Fa18). Prove that a continuous function from $\mathbb{R}$ to $\mathbb{R}$ which maps open sets to open sets must be monotonic.

Problem 13 (Fa84, Fa18). Evaluate $\int_0^{\infty} \frac{x - \sin x}{x^3} dx$.

Problem 14 (Fa78, Fa18). Suppose $h(z)$ is analytic in the whole plane, $h(0) = 3 + 4i$, and $|h(z)| \leq 5$ if $|z| < 1$. What is $h'(0)$?

Problem 15 (Fa95, Fa18, Fa24). Show that an $n \times n$ matrix of complex numbers A satisfying

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|$$

for $1 \leq i \leq n$ must be invertible.

Problem 16 (Fa18). For a real symmetric positive definite matrix A and a vector $v \in \mathbb{R}^n$, show that

$$\int_{\mathbb{R}^n} \exp(-x^T A x + 2v^T x) dx = \frac{\pi^{n/2}}{\sqrt{\det A}} \exp(v^T A^{-1} v)$$

Problem 17 (Fa18). Show that there are no natural numbers $x, y \geq 1$ such that $x^2 + y^2 = 7xy$.

Problem 18 (Fa05, Fa18). Find the smallest n for which S_n contains a cyclic subgroup of order 111.